

Chapter 12

Conic Sections

(Parabola, Ellipse, Hyperbola)

Only One Option Correct Type Questions

1. A hyperbola, having the transverse axis of length $2 \sin \theta$, is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then its equation is [IIT-JEE-2007 (Paper-1)]

(A) $x^2 \operatorname{cosec}^2 \theta - y^2 \sec^2 \theta = 1$ (B) $x^2 \sec^2 \theta - y^2 \operatorname{cosec}^2 \theta = 1$
(C) $x^2 \sin^2 \theta - y^2 \cos^2 \theta = 1$ (D) $x^2 \cos^2 \theta - y^2 \sin^2 \theta = 1$

2. Consider a branch of the hyperbola

$$x^2 - 2y^2 - 2\sqrt{2}x - 4\sqrt{2}y - 6 = 0$$

with vertex at the point A . Let B be one of the end points of its latus rectum. If C is the focus of the hyperbola nearest to the point A , then the area of the triangle ABC is [IIT-JEE-2008 (Paper-2)]

(A) $1 - \sqrt{\frac{2}{3}}$ (B) $\sqrt{\frac{3}{2}} - 1$
(C) $1 + \sqrt{\frac{2}{3}}$ (D) $\sqrt{\frac{3}{2}} + 1$

3. The line passing through the extremity A of the major axis and extremity B of the minor axis of the ellipse $x^2 + 9y^2 = 9$ meets its auxiliary circle at the point M . Then the area of the triangle with vertices at A , M and the origin O is [IIT-JEE-2009 (Paper-1)]

(A) $\frac{31}{10}$ (B) $\frac{29}{10}$
(C) $\frac{21}{10}$ (D) $\frac{27}{10}$

4. The normal at a point P on the ellipse $x^2 + 4y^2 = 16$ meets the x -axis at Q . If M is the mid point of the line segment PQ , then the locus of M intersects the latus rectums of the given ellipse at the points [IIT-JEE-2009 (Paper-2)]

(A) $\left(\pm \frac{3\sqrt{5}}{2}, \pm \frac{2}{7} \right)$ (B) $\left(\pm \frac{3\sqrt{5}}{2}, \pm \frac{\sqrt{19}}{4} \right)$
(C) $\left(\pm 2\sqrt{3}, \pm \frac{1}{7} \right)$ (D) $\left(\pm 2\sqrt{3}, \pm \frac{4\sqrt{3}}{7} \right)$

5. Let (x, y) be any point on the parabola $y^2 = 4x$. Let P be the point that divides the line segment from $(0, 0)$ to (x, y) in the ratio $1 : 3$. Then the locus of P is [IIT-JEE-2011 (Paper-2)]
- (A) $x^2 = y$ (B) $y^2 = 2x$
(C) $y^2 = x$ (D) $x^2 = 2y$
6. Let $P(6, 3)$ be a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If the normal at the point P intersects the x -axis at $(9, 0)$, then the eccentricity of the hyperbola is [IIT-JEE-2011 (Paper-2)]
- (A) $\sqrt{\frac{5}{2}}$ (B) $\sqrt{\frac{3}{2}}$
(C) $\sqrt{2}$ (D) $\sqrt{3}$
7. The ellipse $E_1 : \frac{x^2}{9} + \frac{y^2}{4} = 1$ is inscribed in a rectangle R whose sides are parallel to the coordinate axes. Another ellipse E_2 passing through the point $(0, 4)$ circumscribes the rectangle R . The eccentricity of the ellipse E_2 is [IIT-JEE-2012 (Paper-1)]
- (A) $\frac{\sqrt{2}}{2}$ (B) $\frac{\sqrt{3}}{2}$
(C) $\frac{1}{2}$ (D) $\frac{3}{4}$
8. Let a, b and λ be positive real numbers. Suppose P is an end point of the latus rectum of the parabola $y^2 = 4\lambda x$, and suppose the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through the point P . If the tangents to the parabola and the ellipse at the point P are perpendicular to each other, then the eccentricity of the ellipse is [JEE (Adv)-2020 (Paper-1)]
- (A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$
(C) $\frac{1}{3}$ (D) $\frac{2}{5}$

One or More Option(s) Correct Type Questions

9. Let $P(x_1, y_1)$ and $Q(x_2, y_2)$, $y_1 < 0, y_2 < 0$, be the end points of the latus rectum of the ellipse $x^2 + 4y^2 = 4$. The equations of parabolas with latus rectum PQ are [IIT-JEE-2008 (Paper-1)]
- (A) $x^2 + 2\sqrt{3}y = 3 + \sqrt{3}$ (B) $x^2 - 2\sqrt{3}y = 3 + \sqrt{3}$
(C) $x^2 + 2\sqrt{3}y = 3 - \sqrt{3}$ (D) $x^2 - 2\sqrt{3}y = 3 - \sqrt{3}$
10. An ellipse intersects the hyperbola $2x^2 - 2y^2 = 1$ orthogonally. The eccentricity of the ellipse is reciprocal of that of the hyperbola. If the axes of the ellipse are along the coordinate axes, then [IIT-JEE-2009 (Paper-2)]
- (A) Equation of ellipse is $x^2 + 2y^2 = 2$ (B) The foci of ellipse are $(\pm 1, 0)$
(C) Equation of ellipse is $x^2 + 2y^2 = 4$ (D) The foci of ellipse are $(\pm\sqrt{2}, 0)$

11. The tangent PT and the normal PN to the parabola $y^2 = 4ax$ at a point P on it meet its axis at points T and N , respectively. The locus of the centroid of the triangle PTN is a parabola whose **[IIT-JEE-2009 (Paper-2)]**
- (A) Vertex is $\left(\frac{2a}{3}, 0\right)$ (B) Directrix is $x = 0$
- (C) Latus rectum is $\frac{2a}{3}$ (D) Focus is $(a, 0)$
12. Let A and B be two distinct points on the parabola $y^2 = 4x$. If the axis of the parabola touches a circle of radius r having AB as its diameter, then the slope of the line joining A and B can be **[IIT-JEE-2010 (Paper-1)]**
- (A) $-\frac{1}{r}$ (B) $\frac{1}{r}$
- (C) $\frac{2}{r}$ (D) $-\frac{2}{r}$
13. Let the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be reciprocal to that of the ellipse $x^2 + 4y^2 = 4$. If the hyperbola passes through a focus of the ellipse, then **[IIT-JEE-2011 (Paper-1)]**
- (A) The equation of the hyperbola is $\frac{x^2}{3} - \frac{y^2}{2} = 1$
- (B) A focus of the hyperbola is $(2, 0)$
- (C) The eccentricity of the hyperbola is $\sqrt{\frac{5}{3}}$
- (D) The equation of the hyperbola is $x^2 - 3y^2 = 3$
14. Let L be a normal to the parabola $y^2 = 4x$. If L passes through the point $(9, 6)$, then L is given by **[IIT-JEE-2011 (Paper-2)]**
- (A) $y - x + 3 = 0$ (B) $y + 3x - 33 = 0$
- (C) $y + x - 15 = 0$ (D) $y - 2x + 12 = 0$
15. Tangents are drawn to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$, parallel to the straight line $2x - y = 1$. The point of contact of the tangents on the hyperbola are **[IIT-JEE-2012 (Paper-1)]**
- (A) $\left(\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (B) $\left(-\frac{9}{2\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$
- (C) $(3\sqrt{3}, -2\sqrt{2})$ (D) $(-3\sqrt{3}, 2\sqrt{2})$
16. Let P and Q be distinct points on the parabola $y^2 = 2x$ such that a circle with PQ as diameter passes through the vertex O of the parabola. If P lies in the first quadrant and the area of the triangle ΔOPQ is $3\sqrt{2}$, then which of the following is(are) the coordinates of P ? **[JEE (Adv)-2015 (Paper-1)]**
- (A) $(4, 2\sqrt{2})$ (B) $(9, 3\sqrt{2})$
- (C) $\left(\frac{1}{4}, \frac{1}{\sqrt{2}}\right)$ (D) $(1, \sqrt{2})$

17. Consider the hyperbola $H : x^2 - y^2 = 1$ and a circle S with center $N(x_2, 0)$. Suppose that H and S touch each other at a point $P(x_1, y_1)$ with $x_1 > 1$ and $y_1 > 0$. The common tangent to H and S at P intersects the x -axis at point M . If (l, m) is the centroid of the triangle ΔPMN , then the correct expression(s) is(are)

[JEE (Adv)-2015 (Paper-2)]

- (A) $\frac{dl}{dx_1} = 1 - \frac{1}{3x_1^2}$ for $x_1 > 1$ (B) $\frac{dm}{dx_1} = \frac{x_1}{3(\sqrt{x_1^2 - 1})}$ for $x_1 > 1$
 (C) $\frac{dl}{dx_1} = 1 + \frac{1}{3x_1^2}$ for $x_1 > 1$ (D) $\frac{dm}{dy_1} = \frac{1}{3}$ for $y_1 > 0$

18. Let E_1 and E_2 be two ellipses whose centers are at the origin. The major axes of E_1 and E_2 lie along the x -axis and the y -axis, respectively. Let S be the circle $x^2 + (y - 1)^2 = 2$. The straight line $x + y = 3$ touches the curves S , E_1 and E_2 at P , Q and R respectively. Suppose that $PQ = PR = \frac{2\sqrt{2}}{3}$. If e_1 and e_2 are the eccentricities of E_1 and E_2 , respectively, then the correct expression(s) is (are)

[JEE (Adv)-2015 (Paper-2)]

- (A) $e_1^2 + e_2^2 = \frac{43}{40}$ (B) $e_1 e_2 = \frac{\sqrt{7}}{2\sqrt{10}}$
 (C) $|e_1^2 - e_2^2| = \frac{5}{8}$ (D) $e_1 e_2 = \frac{\sqrt{3}}{4}$

19. Let P be the point on the parabola $y^2 = 4x$ which is at the shortest distance from the center S of the circle $x^2 + y^2 - 4x - 16y + 64 = 0$. Let Q be the point on the circle dividing the line segment SP internally. Then

[JEE (Adv)-2016 (Paper-2)]

- (A) $SP = 2\sqrt{5}$
 (B) $SQ : QP = (\sqrt{5} + 1) : 2$
 (C) The x -intercept of the normal to the parabola at P is 6
 (D) The slope of the tangent to the circle at Q is $\frac{1}{2}$

20. If $2x - y + 1 = 0$ is a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{16} = 1$, then which of the following cannot be sides of a right angled triangle?

[JEE (Adv)-2017 (Paper-1)]

- (A) $2a, 4, 1$ (B) $a, 4, 1$
 (C) $a, 4, 2$ (D) $2a, 8, 1$

21. If a chord, which is not a tangent, of the parabola $y^2 = 16x$ has the equation $2x + y = p$, and midpoint (h, k) , then which of the following is(are) possible value(s) of p, h and k ?

[JEE (Adv)-2017 (Paper-1)]

- (A) $p = 2, h = 3, k = -4$
 (B) $p = 5, h = 4, k = -3$
 (C) $p = -1, h = 1, k = -3$
 (D) $p = -2, h = 2, k = -4$

22. Consider two straight lines, each of which is tangent to both the circle $x^2 + y^2 = \frac{1}{2}$ and the parabola $y^2 = 4x$. Let these lines intersect at the point Q . Consider the ellipse whose center is at the origin $O(0,0)$ and whose semi-major axis is OQ . If the length of the minor axis of this ellipse is $\sqrt{2}$, then which of the following statement(s) is (are) TRUE?
[JEE (Adv)-2018 (Paper-2)]

- (A) For the ellipse, the eccentricity is $\frac{1}{\sqrt{2}}$ and the length of the latus rectum is 1
(B) For the ellipse, the eccentricity is $\frac{1}{2}$ and the length of the latus rectum is $\frac{1}{2}$
(C) The area of the region bounded by the ellipse between the lines $x = \frac{1}{\sqrt{2}}$ and $x = 1$ is $\frac{1}{4\sqrt{2}}(\pi - 2)$
(D) The area of the region bounded by the ellipse between the lines $x = \frac{1}{\sqrt{2}}$ and $x = 1$ is $\frac{1}{16}(\pi - 2)$

23. Define the collections $\{E_1, E_2, E_3, \dots\}$ of ellipses and $\{R_1, R_2, R_3, \dots\}$ of rectangles as follows:

$$E_1 : \frac{x^2}{9} + \frac{y^2}{4} = 1;$$

R_1 : Rectangle of largest area, with sides parallel to the axes, inscribed in E_1 ;

$$E_n : \text{Ellipse } \frac{x^2}{a_n^2} + \frac{y^2}{b_n^2} = 1 \text{ of largest area inscribed in } R_{n-1}, n > 1;$$

R_n : Rectangle of largest area, with sides parallel to the axes, inscribed in $E_n, n > 1$.

Then which of the following options is/are correct ?

[JEE (Adv)-2019 (Paper-1)]

(A) The eccentricities of E_{18} and E_{19} are NOT equal

(B) The length of latus rectum of E_9 is $\frac{1}{6}$

(C) $\sum_{n=1}^N (\text{area of } R_n) < 24$, for each positive integer N

(D) The distance of a focus from the centre in E_9 is $\frac{\sqrt{5}}{32}$

24. Let a and b be positive real numbers such that $a > 1$ and $b < a$. Let P be a point in the first quadrant that

lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Suppose the tangent to the hyperbola at P passes through the point

$(1, 0)$, and suppose the normal to the hyperbola at P cuts off equal intercepts on the coordinate axes. Let Δ denote the area of the triangle formed by the tangent at P , the normal at P and the x -axis. If e denotes the eccentricity of the hyperbola, then which of the following statements is/are TRUE?

[JEE (Adv)-2020 (Paper-2)]

(A) $1 < e < \sqrt{2}$

(B) $\sqrt{2} < e < 2$

(C) $\Delta = a^4$

(D) $\Delta = b^4$

25. Let E denote the parabola $y^2 = 8x$. Let $P = (-2, 4)$, and let Q and Q' be two distinct points on E such that the lines PQ and PQ' are tangents to E . Let F be the focus of E . Then which of the following statements is (are) TRUE ?
[JEE (Adv)-2021 (Paper-2)]

(A) The triangle PFQ is a right-angled triangle

(B) The triangle QPQ' is a right-angled triangle

(C) The distance between P and F is $5\sqrt{2}$

(D) F lies on the line joining Q and Q'

26. Consider the parabola $y^2 = 4x$. Let S be the focus of the parabola. A pair of tangents drawn to the parabola from the point $P = (-2, 1)$ meet the parabola at P_1 and P_2 . Let Q_1 and Q_2 be points on the lines SP_1 and SP_2 respectively such that PQ_1 is perpendicular to SP_1 and PQ_2 is perpendicular to SP_2 . Then, which of the following is/are TRUE ?

[JEE (Adv)-2022 (Paper-1)]

- (A) $SQ_1 = 2$ (B) $Q_1Q_2 = \frac{3\sqrt{10}}{5}$
 (C) $PQ_1 = 3$ (D) $SQ_2 = 1$

Linked Comprehension Type Questions

Paragraph for Q.Nos. 27 to 29

Consider the circle $x^2 + y^2 = 9$ and the parabola $y^2 = 8x$. They intersect at P and Q in the first and the fourth quadrants, respectively. Tangents to the circle at P and Q intersect the x -axis at R and tangents to the parabola at P and Q intersect the x -axis at S .

[IIT-JEE-2007 (Paper-1)]

Choose the correct answer :

27. The ratio of the areas of the triangles PQS and PQR is

- (A) $1 : \sqrt{2}$ (B) $1 : 2$
 (C) $1 : 4$ (D) $1 : 8$

28. The radius of the circumcircle of the triangle PRS is

- (A) 5 (B) $3\sqrt{3}$
 (C) $3\sqrt{2}$ (D) $2\sqrt{3}$

29. The radius of the incircle of the triangle PQR is

- (A) 4 (B) 3
 (C) $\frac{8}{3}$ (D) 2

Paragraph for Q.Nos. 30 and 31

The circle $x^2 + y^2 - 8x = 0$ and hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ intersect at the points A and B .

[IIT-JEE-2010 (Paper-1)]

Choose the correct answer :

30. Equation of a common tangent with positive slope to the circle as well as to the hyperbola is

- (A) $2x - \sqrt{5}y - 20 = 0$
 (B) $2x - \sqrt{5}y + 4 = 0$
 (C) $3x - 4y + 8 = 0$
 (D) $4x - 3y + 4 = 0$

31. Equation of the circle with AB as its diameter is

- (A) $x^2 + y^2 - 12x + 24 = 0$ (B) $x^2 + y^2 + 12x + 24 = 0$
 (C) $x^2 + y^2 + 24x - 12 = 0$ (D) $x^2 + y^2 - 24x - 12 = 0$

Paragraph for Q.Nos. 32 to 34

Tangents are drawn from the point $P(3, 4)$ to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ touching the ellipse at points A and B .

[IIT-JEE-2010 (Paper-2)]

Choose the correct answer :

32. The coordinates of A and B are

(A) $(3, 0)$ and $(0, 2)$

(B) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and $\left(-\frac{9}{5}, \frac{8}{5}\right)$

(C) $\left(-\frac{8}{5}, \frac{2\sqrt{161}}{15}\right)$ and $(0, 2)$

(D) $(3, 0)$ and $\left(-\frac{9}{5}, \frac{8}{5}\right)$

33. The orthocentre of the triangle PAB is

(A) $\left(5, \frac{8}{7}\right)$

(B) $\left(\frac{7}{5}, \frac{25}{8}\right)$

(C) $\left(\frac{11}{5}, \frac{8}{5}\right)$

(D) $\left(\frac{8}{25}, \frac{7}{5}\right)$

34. The equation of the locus of the point whose distances from the point P and the line AB are equal, is

(A) $9x^2 + y^2 - 6xy - 54x - 62y + 241 = 0$

(B) $x^2 + 9y^2 + 6xy - 54x + 62y - 241 = 0$

(C) $9x^2 + 9y^2 - 6xy - 54x - 62y - 241 = 0$

(D) $x^2 + y^2 - 2xy + 27x + 31y - 120 = 0$

Paragraph for Q.Nos. 35 and 36

Let PQ be a focal chord of the parabola $y^2 = 4ax$. The tangents to the parabola at P and Q meet at a point lying on the line $y = 2x + a$, $a > 0$.

[JEE (Adv)-2013 (Paper-2)]

Choose the correct answer :

35. Length of chord PQ is

(A) $7a$

(B) $5a$

(C) $2a$

(D) $3a$

36. If chord PQ subtends an angle θ at the vertex of $y^2 = 4ax$, then $\tan \theta =$

(A) $\frac{2}{3}\sqrt{7}$

(B) $\frac{-2}{3}\sqrt{7}$

(C) $\frac{2}{3}\sqrt{5}$

(D) $\frac{-2}{3}\sqrt{5}$

Paragraph for Q.Nos. 37 and 38

Let a, r, s, t be nonzero real numbers. Let $P(at^2, 2at)$, $Q, R(ar^2, 2ar)$ and $S(as^2, 2as)$ be distinct points on the parabola $y^2 = 4ax$. Suppose that PQ is the focal chord and lines QR and PK are parallel, where K is the point $(2a, 0)$.
[JEE (Adv)-2014 (Paper-2)]

Choose the correct answer :

37. The value of r is

(A) $-\frac{1}{t}$

(B) $\frac{t^2+1}{t}$

(C) $\frac{1}{t}$

(D) $\frac{t^2-1}{t}$

38. If $st = 1$, then the tangent at P and the normal at S to the parabola meet at a point whose ordinate is

(A) $\frac{(t^2+1)^2}{2t^3}$

(B) $\frac{a(t^2+1)^2}{2t^3}$

(C) $\frac{a(t^2+1)^2}{t^3}$

(D) $\frac{a(t^2+2)^2}{t^3}$

Paragraph for Q.Nos. 39 and 40

Let $F_1(x_1, 0)$ and $F_2(x_2, 0)$, for $x_1 < 0$ and $x_2 > 0$, be the foci of the ellipse $\frac{x^2}{9} + \frac{y^2}{8} = 1$. Suppose a parabola having vertex at the origin and focus at F_2 intersects the ellipse at point M in the first quadrant and at point N in the fourth quadrant.
[JEE (Adv)-2016 (Paper-2)]

Choose the correct answer :

39. The orthocentre of the triangle F_1MN is

(A) $\left(-\frac{9}{10}, 0\right)$

(B) $\left(\frac{2}{3}, 0\right)$

(C) $\left(\frac{9}{10}, 0\right)$

(D) $\left(\frac{2}{3}, \sqrt{6}\right)$

40. If the tangents to the ellipse at M and N meet at R and the normal to the parabola at M meets the x -axis at Q , then the ratio of area of the triangle MQR to area of the quadrilateral MF_1NF_2 is

(A) 3 : 4

(B) 4 : 5

(C) 5 : 8

(D) 2 : 3

Matrix-Match / Matrix Based Type Questions

41. Match the statements in **Column-I** with the properties in **Column-II**. [IIT-JEE-2007 (Paper-2)]

Column-I**Column-II**

- | | |
|--|----------------------------------|
| (A) Two intersecting circles | (p) Have a common tangent |
| (B) Two mutually external circles | (q) Have a common normal |
| (C) Two circles, one strictly inside the other | (r) Do not have a common tangent |
| (D) Two branches of a hyperbola | (s) Do not have a common normal |

42. Match the conics in **Column-I** with the statements/expressions in **Column-II**. [IIT-JEE-2009 (Paper-1)]

Column-I**Column-II**

- | | |
|---------------|--|
| (A) Circle | (p) The locus of the point (h, k) for which the line $hx + ky = 1$ touches the circle $x^2 + y^2 = 4$ |
| (B) Parabola | (q) Points z in the complex plane satisfying $ z + 2 - z - 2 = \pm 3$ |
| (C) Ellipse | (r) Points of the conic have parametric representation $x = \sqrt{3} \left(\frac{1-t^2}{1+t^2} \right), y = \frac{2t}{1+t^2}$ |
| (D) Hyperbola | (s) The eccentricity of the conic lies in the interval $1 \leq e < \infty$ |
| | (t) Points z in the complex plane satisfying $\operatorname{Re}(z + 1)^2 = z ^2 + 1$ |

43. A line $L : y = mx + 3$ meets y -axis at $E(0, 3)$ and the arc of the parabola $y^2 = 16x$, $0 \leq y \leq 6$ at the point $F(x_0, y_0)$. The tangent to the parabola at $F(x_0, y_0)$ intersects the y -axis at $G(0, y_1)$. The slope m of the line L is chosen such that the area of the triangle EFG has a local maximum. [JEE (Adv)-2013 (Paper-2)]

Match Column-I with Column-II and select the correct answer using the code given below the lists:

Column-I**Column -II**

- | | |
|---------------------------------------|------------------|
| P. $m =$ | 1. $\frac{1}{2}$ |
| Q. Maximum area of $\triangle EFG$ is | 2. 4 |
| R. $y_0 =$ | 3. 2 |
| S. $y_1 =$ | 4. 1 |

Codes :

- | | | | |
|-------|---|---|---|
| P | Q | R | S |
| (A) 4 | 1 | 2 | 3 |
| (B) 3 | 4 | 1 | 2 |
| (C) 1 | 3 | 2 | 4 |
| (D) 1 | 3 | 4 | 2 |

Answer Q.43, Q.44 and Q.45 by appropriately matching the information given in the three columns of the following table.

Column 1, 2, and 3 contain conics, equations of tangents to the conics and points of contact, respectively.		
Column 1	Column 2	Column 3
(I) $x^2 + y^2 = a^2$	(i) $my = m^2x + a$	(P) $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$
(II) $x^2 + a^2y^2 = a^2$	(ii) $y = mx + a\sqrt{m^2 + 1}$	(Q) $\left(\frac{-ma}{\sqrt{m^2 + 1}}, \frac{a}{\sqrt{m^2 + 1}}\right)$
(III) $y^2 = 4ax$	(iii) $y = mx + \sqrt{a^2m^2 - 1}$	(R) $\left(\frac{-a^2m}{\sqrt{a^2m^2 + 1}}, \frac{1}{\sqrt{a^2m^2 + 1}}\right)$
(IV) $x^2 - a^2y^2 = a^2$	(iv) $y = mx + \sqrt{a^2m^2 + 1}$	(S) $\left(\frac{-a^2m}{\sqrt{a^2m^2 - 1}}, \frac{-1}{\sqrt{a^2m^2 - 1}}\right)$

44. The tangent to a suitable conic (Column 1) at $\left(\sqrt{3}, \frac{1}{2}\right)$ is found to be $\sqrt{3}x + 2y = 4$, then which of the following options is the only correct combination? **[JEE (Adv)-2017 (Paper-1)]**

- (A) (IV) (iii) (S) (B) (II) (iii) (R)
(C) (IV) (iv) (S) (D) (II) (iv) (R)

45. If a tangent to a suitable conic (Column 1) is found to be $y = x + 8$ and its point of contact is $(8, 16)$, then which of the following options is the only correct combination? **[JEE (Adv)-2017 (Paper-1)]**

- (A) (III) (i) (P) (B) (I) (ii) (Q)
(C) (II) (iv) (R) (D) (III) (ii) (Q)

46. For $a = \sqrt{2}$, if a tangent is drawn to a suitable conic (Column 1) at the point of contact $(-1, 1)$, then which of the following options is the only correct combination for obtaining its equation? **[JEE (Adv)-2017 (Paper-1)]**

- (A) (II) (ii) (Q) (B) (I) (i) (P)
(C) (I) (ii) (Q) (D) (III) (i) (P)

47. Let $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where $a > b > 0$, be a hyperbola in the xy -plane whose conjugate axis LM subtends an angle of 60° at one of its vertices N . Let the area of the triangle LMN be $4\sqrt{3}$. **[JEE (Adv)-2018 (Paper-2)]**

List-I

- P. The length of the conjugate axis of H is
Q. The eccentricity of H is
R. The distance between the foci of H is
S. The length of the latus rectum of H is

List-II

1. 8
2. $\frac{4}{\sqrt{3}}$
3. $\frac{2}{\sqrt{3}}$
4. 4

The correct option is

- (A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$ (B) $P \rightarrow 4; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 2$
(C) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 2$ (D) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

48. Consider the ellipse

$$\frac{x^2}{4} + \frac{y^2}{3} = 1$$

Let $H(\alpha, 0)$, $0 < \alpha < 2$, be a point. A straight line drawn through H parallel to the y -axis crosses the ellipse and its auxiliary circle at points E and F respectively, in the first quadrant. The tangent to the ellipse at the point E intersects the positive x -axis at a point G . Suppose the straight line joining F and the origin makes an angle ϕ with the positive x -axis.

[JEE (Adv)-2022 (Paper-1)]

List-I

List-II

(i) If $\phi = \frac{\pi}{4}$, then the area of the triangle FGH is

(P) $\frac{(\sqrt{3}-1)^4}{8}$

(ii) If $\phi = \frac{\pi}{3}$, then the area of the triangle FGH is

(Q) 1

(iii) If $\phi = \frac{\pi}{6}$, then the area of the triangle FGH is

(R) $\frac{3}{4}$

(iv) If $\phi = \frac{\pi}{12}$, then the area of the triangle FGH is

(S) $\frac{1}{2\sqrt{3}}$

(T) $\frac{3\sqrt{3}}{2}$

The correct option is:

(A) (I) $\rightarrow$ (R); (II) $\rightarrow$ (S); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (P)

(B) (I) $\rightarrow$ (R); (II) $\rightarrow$ (T); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (P)

(C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (T); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (P)

(D) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (Q); (IV) $\rightarrow$ (P)

Integer / Numerical Value Type Questions

49. The line $2x + y = 1$ is tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If this line passes through the point of intersection of the nearest directrix and the x -axis, then the eccentricity of the hyperbola is [IIT-JEE-2010 (Paper-1)]

50. Consider the parabola $y^2 = 8x$. Let Δ_1 be the area of the triangle formed by the end points of its latus rectum and the point $P\left(\frac{1}{2}, 2\right)$ on the parabola, and Δ_2 be the area of the triangle formed by drawing tangents at P

and at the end points of the latus rectum. Then $\frac{\Delta_1}{\Delta_2}$ is

[IIT-JEE-2011 (Paper-1)]

51. Let S be the focus of the parabola $y^2 = 8x$ and let PQ be the common chord of the circle $x^2 + y^2 - 2x - 4y = 0$ and the given parabola. The area of the triangle PQS is [IIT-JEE-2012 (Paper-1)]

52. A vertical line passing through the point $(h, 0)$ intersects the ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$ at the points P and Q . Let the tangents to the ellipse at P and Q meet at the point R . If $\Delta(h)$ = area of the triangle PQR ,

$$\Delta_1 = \max_{1/2 \leq h \leq 1} \Delta(h) \text{ and } \Delta_2 = \min_{1/2 \leq h \leq 1} \Delta(h), \text{ then } \frac{8}{\sqrt{5}} \Delta_1 - 8 \Delta_2 =$$

[JEE (Adv)-2013 (Paper-1)]

53. If the normals of the parabola $y^2 = 4x$ drawn at the end points of its latus rectum are tangents to the circle $(x-3)^2 + (y+2)^2 = r^2$, then the value of r^2 is [JEE (Adv)-2015 (Paper-1)]

54. Let the curve C be the mirror image of the parabola $y^2 = 4x$ with respect to the line $x + y + 4 = 0$. If A and B are the points of intersection of C with the line $y = -5$, then the distance between A and B is

[JEE (Adv)-2015 (Paper-1)]

55. Suppose that the foci of the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$ are $(f_1, 0)$ and $(f_2, 0)$ where $f_1 > 0$ and $f_2 < 0$. Let P_1 and P_2 be two parabolas with a common vertex at $(0, 0)$ and with foci at $(f_1, 0)$ and $(2f_2, 0)$, respectively. Let T_1 be a tangent to P_1 which passes through $(2f_2, 0)$ and T_2 be a tangent to P_2 which passes through $(f_1, 0)$. If m_1 is the slope of T_1 and m_2 is the slope of T_2 , then the value of $\left(\frac{1}{m_1^2} + m_2^2\right)$ is **[JEE (Adv)-2015 (Paper-2)]**

Question Stem for Question Nos. 56 and 58

Consider the region $R = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \geq 0 \text{ and } y^2 \leq 4 - x\}$. Let F be the family of all circles that are contained in R and have centers on the x -axis. Let C be the circle that has largest radius among the circles in F . Let (α, β) be a point where the circle C meets the curve $y^2 = 4 - x$. **[JEE (Adv)-2021 (Paper-2)]**

56. The radius of the circle C is _____.

57. The value of α is _____.

58. Let E be the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$. For any three distinct points P , Q and Q' on E , let $M(P, Q)$ be the mid-point of the line segment joining P and Q , and $M(P, Q')$ be the mid-point of the line segment joining P and Q' . Then the maximum possible value of the distance between $M(P, Q)$ and $M(P, Q')$, as P , Q and Q' vary on E , is _____. **[JEE (Adv)-2021 (Paper-2)]**

59. Consider the hyperbola $\frac{x^2}{100} - \frac{y^2}{64} = 1$ with foci at S and S_1 , where S lies on the positive x -axis. Let P be a point on the hyperbola, in the first quadrant. Let $\angle SPS_1 = \alpha$, with $\alpha < \frac{\pi}{2}$. The straight line passing through the point S and having the same slope as that of the tangent at P to the hyperbola, intersects the straight line S_1P at P_1 . Let δ be the distance of P from the straight line SP_1 , and $\beta = S_1P$. Then the greatest integer less than or equal to $\frac{\beta\delta}{9} \sin \frac{\alpha}{2}$ is _____. **[JEE (Adv)-2022 (Paper-2)]**

